Multi-Wavelength Photoplethysmographic Sensors: innovations for non-invasive physiological parameters monitoring

Ferretti Samuel, Ghezzi Martino, Gumiero Chiara, Magni Francesca, and Merlo Francesco

***Abstract*—Multi-wavelength photoplethysmography is a technique that measures the absorption of visible and near-infrared light by blood vessels and other biological tissues. These measurements can be converted into various clinically important biological information. The most studied and accredited over the years are heart rate, oxygen saturation, and blood pressure. However, photoplethysmography has the potential to measure other clinically relevant data, as will be discussed and analyzed in this study. This review article provides a comprehensive overview of recent advancements of multi-wavelength photoplethysmographic sensors and their innovative applications in areas like microcirculation assessment, non-invasive glucose monitoring, and improved drug delivery monitoring, as well as classical photoplethysmography enhancements to improve the measurement of standard parameters.**

The review discusses the underlying principles for each topic and explores relevant works that have addressed their application. A special focus has been placed on analyzing the choice of wavelengths in multi-channel photoplethysmographic systems and their relevance to the measured parameter. Lastly, our work highlights the challenges and limitations of current multi-wavelength photoplethysmographic systems while also providing recommendations for future research directions.

***Index Terms*—Blood glucose, drug delivery, microcirculation, motion artifacts, multi-wavelength photoplethysmography, oxygen saturation, physiological monitoring.**

I. Introduction

THE current approach to treating chronic diseases is transitioning towards employing cost-effective continuous monitoring tools. These tools primarily aim to bolster the early prevention, diagnosis, and treatment of pathologies [1]. The World Health Organization (WHO) supports continuous monitoring as a key way to evaluate physiological conditions, track disease progression, and facilitate everyday health management independently [2]. Consequently, there's a pressing need for user-friendly systems devoid of complex technical knowledge [2]. Moreover, it's imperative to ensure that monitoring devices integrate into patients' daily routines without causing inconvenience [3]. In response to these requisites, the utilization of photoplethysmographic (PPG) sensors has garnered considerable attention as they provide a non-invasive and accessible method for continuous health monitoring.

A standard PPG device comprises a light source and a photodetector, enabling a non-invasive evaluation of blood volume changes in the microcirculation of a reachable body part, such as the fingertip pulp. PPG is able to estimate the amount of light absorbed or reflected by the blood flowing within blood vessels in a pulsatile manner.

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As the heart contracts, the rapid surge in blood flow volume within the vessels diminishes the intensity of the light emitted by the PPG measuring device, enabling it to perceive variations in blood flow volume throughout the cardiac cycle. The PPG signal exhibits three characteristic events: the dicrotic notch, the systolic peak, and the diastolic peak. The dicrotic notch is caused by the closure of the aortic valve, marking the end of the systolic phase and the beginning of the diastolic phase. The systolic and diastolic peaks, on the other hand, respectively indicate the perception of ventricular contraction and relaxation at the point of measurement (Fig. 1.). The Pulse Arrival Time (PAT) is defined as the time taken from the pulse waveform to traverse from the heart to the point in which the measurement takes place. As we can see in Fig. 1, the PPG waveform consists of pulsatile (AC) and non-pulsatile (DC) components. The AC component offers insights into volumetric fluctuations in blood vessels, correlating with heart rate. Meanwhile, the DC component gauges light absorption by tissue, veins and blood at the measurement

site, indicating changes in blood vessel volume capacity, such as those influenced by respiration [4],[5].

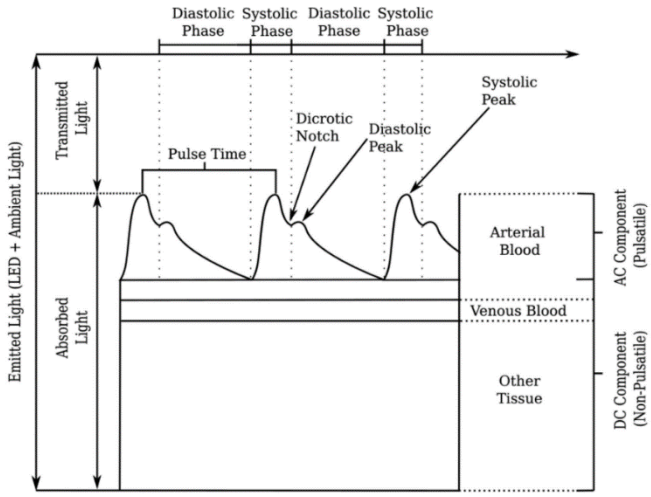

Fig. 1. A PPG waveform adapted from Lemay *et al.* [6].

When referring to multi-wavelength photoplethysmography (MWPPG), it denotes a technique that utilizes multiple light sources within the sensor. The efficacy of MWPPG is based on the blood's property to absorb light differently at various wavelengths, allowing for detailed insights into blood flow and physiological parameters. The advantages of using MWPPG include:

- Improved blood component discrimination. Different wavelengths assist in distinguishing, for example, between oxygenated and deoxygenated hemoglobin, enhancing the ability to monitor blood oxygen levels (SpO2) and related parameters [2].
- Improved resilience to motion artifacts and variations in skin melanin content and skin temperature. In this way, it is possible to provide robust estimations of physiological parameters [2].

This study is aimed to investigate innovative applications of MWPPG in monitoring blood glucose levels, drug delivery and microcirculation: areas that have recently gained interest but are not yet extensively covered in the literature due to their novel nature.

We also revisited well-studied parameters such as heart rate, blood pressure, vascular aging and respiratory physiology, analyzing emerging strategies aimed at overcoming existing challenges. Finally, this research examines the various wavelengths utilized within PPG sensors to assess the accuracy of these devices in monitoring physiological parameters compared to gold standard references.

II. Materials And Methods

The papers that constitute this review were retrieved from the following search engines and digital libraries: PubMed, Scopus, Google Scholar, and Institute of Electrical and Electronics Engineers (IEEE) Xplore Digital Library. The research included articles featuring the following combinations of keywords within the title or abstract:

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[Multi-Wavelength AND (Photoplethysmographic OR PPG) AND Sensor*]
AND
[Wavelength AND PPG AND Physiological Parameters]
AND
[Photoplethysmographic AND Sensors AND (Health care OR Applications)]
AND
[(Photoplethysmography OR PPG) AND Glucose]
AND
[Non-invasive AND PPG AND Glucose]
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We expanded the collection of analyzed papers by including additional articles found in the references of the identified papers from the above-mentioned searches. In the initial stages of the research 7 main applications of multiwavelength PPG in innovative non-invasive monitoring of patient physiological parameters were identified. These were: blood glucose level monitoring, hypertension monitoring, respiratory problems monitoring, sympathetic system monitoring, microcirculation monitoring, drug distribution monitoring and innovative approaches in the monitoring of standard parameters. Of these 7, a further selection was carried out which led us to retain applications that exhibited the following characteristics:

- A minimum of 5 papers related to that specific application could be identified in the literature.
- Comprehensive reviews addressing the considered application were lacking in the literature.

In this way, the 4 applications of MWPPG that are the subject of this review have been identified.

III. Results

As a result of the literature search described in Materials and Methods section, we reviewed in detail a total of 25 articles, divided as follows: 6 articles on blood glucose level monitoring, 5 articles on drug distribution monitoring within the body, 6 articles on microcirculation monitoring, and 8 articles on the main innovations and recently employed technologies in monitoring standard physiological parameters.

A. Drug Delivery Monitoring

MWPPG can be highly beneficial in pharmacological therapies by detecting drug concentration in the blood in real time [7]. This task is very important as many substances, injected in patients for specific treatments, have significant inter-individual variability in bioavailability and toxicity:

some subjects do not achieve the desired exposure, while others experience adverse side effects due to over-exposure [7]. This is what can happen if the management of pharmacotherapy is merely estimated according to the available dosing guidelines, without accounting for the high variability of individual differences [7].

The functioning principle is based on the resolution of a system of equations derived from a manipulation of the Lambert-Beer equation [8]. ΔA is the ratio between the blood pulsatile components (AC) and the non-pulsatile components (DC): with a multi-wavelength approach we can examine R which is the ratio between ΔA for different couples of wavelengths (1). Then, by performing the previous cited manipulations, it's possible to achieve formulas that are dependent only on the effective attenuation coefficients of reduced (μHb) and oxygenated ($\mu HbO2$) whole blood and on the drug's optical density ($\mu drug$) (2). At the end, by solving the system of equations for $\mu drug$ and S (saturation) by using different pairs of wavelengths, the concentration of the drug can be calculated [8].

The results obtained by using this technique have been tested in different experiments that involved different therapeutic nanoparticles and drugs that could be toxic in excessive concentrations: gold nanorods used in detection and treatment of cancer (extinction peak at ~805nm) [8], anti-malarial quinine (extinction peak at ~350nm) [9] and anti-fungal agent amphotericin b (extinction peak at ~355nm) [10]. Equations (1) and (2) are an example for quinine, considering the values at 340nm and 940nm [9].

$$R_{340/940} = \frac{V_{AC340}}{V_{DC340}} \Big/ \frac{V_{AC940}}{V_{DC940}} \quad (1)$$

$$R_{340/940} = \frac{S(\mu_{HbO2}^{340}) + (1 - S)(\mu_{Hb}^{340}) + \mu_{quinine}^{340}}{S(\mu_{HbO2}^{940}) + (1 - S)(\mu_{Hb}^{940}) + \mu_{quinine}^{940}} \quad (2)$$

The optical densities and concentrations were calculated repeatedly at regular intervals after the injection and were compared with the reference values from UV/Vis analysis of blood (which was used as gold standard) taken at the same

TABLE I
DETAILS OF MWPPG STUDIES ON DRUG DELIVERY

	Measured drug	Number of subjects	Sensor position	Measurement condition	Reference Signal
Gregory J. Michalak et al., 2010 [8]	Gold nanorods	5 mice	Mouse tail	Injection of 4.5 µl/g of gold nanorods	Spectrometric UV/Vis blood analysis
Pratik Adhikari et al., 2014 [9]	Quinine			Injection of 80 µL quinine dihydrochloride solution (3.23µg/mL of Quinine)	
Pratik Adhikari et al., 2015 [10]	Amphotericin b			Injection of 5mg/kg solution at 5mg/mL Amphotericin b concentration	

Details of the different studies on the drug delivery, underlying which particle was analyzed, which were the experiments' settings, and which was the gold standard considered.

TABLE II
RESULTS OF MWPPG STUDIES ON DRUG DELIVERY

	Wavelengths	Reference Signal	Correlation coefficient	Precision Bland-Altman	Bias Bland-Altman
Gregory J. Michalak et al., 2010 [8]	660nm 805nm 940nm	Spectrometric UV/Vis blood analysis	r^2 = 0.94	± 0.86 OD	-0.22 OD
Pratik Adhikari et al., 2014 [9]	340nm 660nm 940nm		r^2 = 0.96	± 0.56 OD	+0.17 OD
Pratik Adhikari et al., 2015 [10]	355nm 660nm 940nm		r^2 = 0.88	± 0.52 OD	+0.23 OD

Summary of the specific results obtained in the different studies including also the wavelength used, as well as the gold standard to which the statistics refer. Some of these data are summed up in [12]. OD = Optical Density

time points; also Bland Altman analysis was performed to assess the bias and the precision between the results of both techniques [11]. The calculated optical densities of each circulating agent showed a linear fit with the ones obtained with UV/Vis blood analysis and in particular:

- The gold nanorods have a linear fit of y = 0.87x+0.09 [11].
- The quinine has a linear fit of y = 0.927x+0.30 [11].
- The amphotericin b has a linear fit of y = 0.85x+0.41 [11].

The main experiments' features and results are showed in Table I and Table II.

B. Microcirculation

The terminal part of the systemic circulation, which is composed of capillaries, venules and arterioles [12], could be an interesting disease detector; in fact, the fluctuations in blood parameters at this level are non-negligible signs that are able to anticipate the presence of imminent issues concerning the entire circulatory system [13].

We are not only talking about cardiovascular diseases such as diabetes mellitus, ischemic cardiomyopathies, obesity, and stroke, but also the neuro-immune-endocrine ones such as neuropathy or Raynaud's syndrome. Also, the fact that microcirculation could change as a consequence of external actions is supposed to reveal nerve injury or should be used to monitor the regional anesthesia [14].

When the oxygenation of an organ is compromised, the perfusion of its tissue must compensate for the lack of oxygen; that's why perfusion is considered a key parameter and, since it's often accompanied with variations in blood pressure, also hypertension is investigated.

For this reason, in the Table III we can easily find the PI, perfusion index, which is preferred to be expressed as percentage changes from the baseline since some controlled perturbations may be introduced, for example, consider the anesthesia injection, the application of a vasodilator in the liniment test, the heat and cold tests or the application of a local pressure.

The pulsatile component (AC) and the slow one (DC) extracted from the PPG signal can be combined to obtain the PI, equation (3):

$$PI = \frac{100\sqrt{\frac{\sum_{i=1}^{N} pi^2}{N}}}{\frac{1}{N}\sum_{i=1}^{N} DCi} \tag{3}$$

where N is the number of samples considered to calculate the moving average of DC, pi is a PPG AC signal in the i'th sample, while multiplying this formula by a factor of 100, as shown, allows us to have the percentage [15]. Other expressions could be introduced to measure the skin microcirculation PI indirectly; there is a relationship between PPG and blood volume "Vol(t)", in first approximation in equation (4) we have:

$$Q = \frac{d(Vol(t))}{dt} = \frac{d(PPG)}{dt} \tag{4}$$

where Q is the blood flow [13].

It's also possible to implement an algorithm capable of extracting data from the NIR rPPG signal (we have a remote video camera and LED ring) in order to create skin microcirculation distribution maps that could show the changes in the skin perfusion after administration of the local anesthetics [14].

As anticipated, also blood pressure BP is a parameter of interest [16], as well as the mean arterial pressure MAP or again blood pulsations amplitude BPA. The latter is related to systolic and diastolic pressure [17], the idea is to understand if its value could be extracted from PPG signal calculating AC/DC ratio during body reaction on various stimuli such as cooling test. Have a look at Table IV.

TABLE III
DETAILS OF MWPPG STUDIES ON MICROCIRCULATION

	Estimated Parameters	Number of subjects	Sensor position	Measurement condition
Sirkia et al., 2023 [16]	BP, MAP	10	Finger	A ramp of pressure (2 min)
	DC component	19		PIV, a ramp of pressure is applied for 2 min then held constant and finally released
Rubins et al., 2016 [14]	Variations of backscattered illumination due to blood parameters (HR, spO2, PI)	6 patients who are about to undergo hand surgery	15 cm from patient's hand	RA 1 min before the injection and in the following 9 minutes from this event a video is recorded

Acevedo et al., 2023 [13]	15 features (including PI)	2 different samples from 31 healthy subjects	Finger	90 sec at rest, then application of pressure for 90 sec
			Tongue	2 min of acquisition at rest
Kamshilin et al., 2016 [17]	The delay related to the increase of BPA	9 healthy subjects	Distance between hand's palm and the LEDs and camera is 21,5 cm	The finger is cooled down by immersing into ice water (T=5°C) for 60 seconds, the acquisition is held in the following 2 minutes
Marcinkevics et al., 2016 [15]	PI	10 Caucasian subjects	15 cm from the patient's hand	Liniment test
		10 healthy subjects		Heating test T=40°C
		10 healthy subjects		RA
		7 patients who are about to undergo hand surgery		Post occlusive reactive hyperemia test

Presentation of 5 of the 6 papers with their estimated parameters, their number of subjects, where the sensor is located, the reference signal and the measurement protocol. PI=perfusion index, BPA= amplitude of blood pulses=AC/DC, HR=heart rate, MAP=mean arterial pressure, spO2=peripheral oxygen saturation, RA=regional anesthesia.

TABLE IV
DETAILS OF MWPPG STUDIES ON MICROCIRCULATION

	Wavelengths	Reference Signal	Additional Evidence	Correlation Coefficient (r)
Sirkia et al., 2023 [16]	465nm 515nm 590nm 640nm 880nm	Brachial cuff BP (Omron M3)		r=0,934 between infrared channel and the pressure one (obtained through inference of information) r=0,994 between infrared and red channels r=0,804 between pressure channel and the reference device (brachial cuff)
			DC component of the yellow channel shows the highest variation	
Rubins et al., 2016 [14]	760nm rPPG	Thermal images of the hand	Efficiency in monitoring blood volume changes	
Acevedo et al., 2023 [13]	Red, infrared, green (between 470 and 550 nm)	Comparison between signals themselves	$IQRnopressgreen$=2,046 % p<0.001	
			$IQRpressgreen$=0,1097 % p<0.001	
			$IQRnopressgreen$=1,253 %	
Kamshilin et al., 2016 [17]	525nm 660nm 880nm	Comparison between signals themselves	BPA=BPAaverage±26% more than one trial per subject	r>0,78 p<0,05 of DC and AC components and the growth rate of BPA
Marcinkevics et al., 2016 [15]	530nm 810nm	LDI	$iPPG530nm$=370±136% $iPPG810nm$=123± 29% LDI=530±327%	r= 0.98±0.01 (p<0.001) fra LDI e $iPPG530nm$
			$iPPG530nm$=659±297% $iPPG810nm$=99± 52% LDI=611±228%	r=0.97±0.01 (p<0.001) fra LDI e $iPPG530nm$
		LDI & comparison between signals themselves	$iPPG530nm$=94±24% $iPPG810nm$=71± 23%	
		Comparison between signals themselves	$iPPG530nm=413\pm80$ % $iPPG810nm=359\pm191\%$	

Reported wavelengths, reference signal, other types of results and the value of correlation if available. rPPG= remote PPG, IQR=interquantile range, LDI=Laser Doppler Imaging.

C. Blood Glucose Monitoring

Diabetes is a very diffused disease and if its prevalence was around 171 million of people in 2004, in 2030 will affect 366 million of people of all age groups, even if other studies hyphotesize larger numbers. This disease depends on the fact that the body fails to deal with insuline. Individual with diabetes can suffer of long-term complications, such as retinopathy, cardiovascular diseases and kidney diseases. There is no cure, so preventive medicine can be useful for the detection of diabetes and for patients suffering this disease who need to be daily monitored

The classical solution to monitor diabetes, like glucometers, consist in analyzing the glucose level of a patient, but these methods, still very diffused, are invasive and painful measurements, taking blood sampling using finger-prick. Measurements provided by glucometers are expensive because glucose strip can be used only once and, furthermore, this technique does not report concentration changes in real time. Nowadays exist examples of devices based on minimally invasive systems. These implantable sensors are commercialized but have a limited lifetime, have some delays in reading and suffer stability problems. In recent years PPG is starting to represent a necessary alternative, the diffusion of non-invasive devices is preferable also taking count of the size of the population with this disease, because these instruments could be integrated also inside smartwatches. Non-invasive devices can be based on acoustic waves, microwaves, bioelectric impedance or PPG.

The [18] study demonstrates that a BG biosensor based on 4 LEDs on Vis-Nir spectrum (950, 860, 660, 530 nm) has better performances in estimating BG levels than a 2 LEDs NIR-feature based model and a 2 LEDs VIS-feature model. As a reference it's been used a finger-pricked BG device HealthPro, measurements have been taken on 12 subjects. The calibration model used is PLS (partial least square algorithm) and the obtained the correlation coefficient Rp=0.86. While [19] study focus on estimating HbA1c levels, a quantitative parameter obtainable through MW-PPG. In this paper the main theme consists in the comparison between a 3-wavelength model and a 2-wavelength model to understand which works better and if there are significative differences in assessing values. The selected wavelengths are 465, 525, 615 nm. It's been demonstrated that the 3-wavelength model works better and the error in HbA1c estimation is around 10^{-14} %, compared to 2-wavelength model whose error is in the range of –1.306% to 0.047%. The 2-wavelength model has a better signal to noise ratio with respect to the 3-wavelength model, but SpO2 levels can significantly influences the accuracy.

Nowadays the most diffused methods to study BG levels takes the measures on the ear or on the fingertip, but for diabetes patients it's certainly more comfortable applying the non-invasive on the wrist, as [20] study does, instead of the finger. In this case the PPG sensor works in reflection mode on the radial arteries and requires a Vis-Nir PPG, because the light needs to penetrate several skin layers.

It's possible to apply machine learning techniques in support of PPG data for the estimation of BG levels. In [20] are tested different algorithms, which have been provided with 2 different data: PPG signal and IMF-based features. The selected LEDs have wavelengths of 465,525,615 nm and as reference it's used a commercial device: CareSens II blood glucometer. The best performances are obtained using the Catboost machine learning model and considering 50 features (wavelength-based features and IMF-based top features).

In these conditions the Pearson's r is 0.96 and R^2=0.92. Different results are obtained in [21], where the best performances are obtained with Random Forest and R^2=0.9, in fact the value of coefficient of determination is used in this study for the selection of the ML algorithm. This study has worked on 410 subjects, the features selected basing on selected physiological facts.

The [22] instead, for the estimation of BG level, has found a correlation equation between blood glucose and digital output data. The BG level, in this case, is negatively correlated with output r=-0.81. The results show that this system, appliable on the finger of the patient, has great performances in the range of 75-130 mg/dl (R^2=0.99), but overall results are less precise and show that in the total range of 75-289 mg/dl (R^2=0.68). So it could provide errors in measuring some cases of hyperglycemia. The results of the studies just described are summarized in Table V.

It's possible also to consider a system based on a neural network (BPNN) with a 2 wavelength PPG (660nm, 900nm) and a sensor measuring bioelectric impedance positioned under the wrist [23]. As input to BPNN were given 30 features (statistical, physiological and impedance features) and the reported R^2= 0.997. Clarke error grid is typically used to quantify the accuracy in estimating the blood glucose level. The BPNN-based system and the Catboost-based report that all the pointes measured are in region A of the Clarke error grid. While [21] study has 87.7% of points in region A, 10.3% in region B and 2% in region D but 0% in region E so cases of hyperglycemia are not confused with hypoglycemia and vice versa. The results of the studies just described are summarized in Table VI.

TABLE V

DETAILS OF MWPPG STUDIES ON BLOOD GLUCOSE MONITORING

	Wavelengths	Number of subjects	Sensor position	Reference Signal	Correlation (values referred to glucose)
Vega Pradana Rachim et al., 2019 [18]	2 LEDs VIS 530nm 660nm 2 LEDs Nir 860nm 950nm 4 LEDs VIS-Nir 950nm 860nm 660nm 530nm	12	Wrist	The finger-pricked BG device HealthPro	Rp: VIS=0.55 NIR=0.67 VIS-NIR=0.86
Shifat Hossain et al., 2021 [19]	465nm 525nm 615nm	Not reported	Not reported	To study noise levels in Hb1Ac estimation model: AFE4404-based sensor and TCS34725-based sensor	3-wavelength model: error 10^-14 %. 2-wavelength model: error in the range -1.306% to 0.047% of HbA1c
Shama Satter et al., 2021 [20]	465nm 525nm 615nm	34	Wrist	CareSens II blood glucometer	R^2=0.92 Rp=0.96
Enric Monte-Moreno, 2011 [21]	2 LEDs Red Nir	410	Fingertip	BGL: Blood glucometer Accu-Chek Aviva SBP and DBP: aneroid sphygmomanometer	Best performances: Random Forest R^2=0.9
Preya Anupongongarch et al., 2022 [22]	2 LEDs 950nm 630nm	300	Fingertip	Blood test at the hospital in the same day	r = -0.81 (p<0.001) Between 75-130 mg/dL: R^2=0.99. Between 75-289 mg/dL: R^2=0.68. Paired t-test p=0.135

Reported number of tests, wavelengths, sensor placement, reference signal and value of correlation. LEDs = light emitting diodes, VIS = visible, Nir = near infrared, BG = blood glucose, Rp = correlation coefficient, Hb1Ac = glycated hemoglobin, BGL = blood glucose level, SBP = systolic blood pressure, DBP = diastolic blood pressure.

TABLE VI
STUDIES ON BLOOD GLUCOSE MONITORING AND CLARKE ERROR GRID

	Purpose	Clarke Error Grid
Vega Pradana Rachim et al., 2019 [18]	Studying indirectly BG concentration from the pulsatile and non-pulsatile properties of the PPG signal. Calibration obtained through partial least square algorithm	Not reported
Shifat Hossain et al., 2021 [19]	To quantify glycated hemoglobin (HbA1c) level using a MWPPG system	Not reported
Shama Satter et al., 2021 [20]	Comparison of machine learning systems trained to recognize blood glucose. The study uses data extracted from signal before and after EMD	With Catboost all points in A
Enric Monte-Moreno, 2011 [21]	BGL, SBP and DBP calculation by evaluating effects on PPG of physiological alterations and information on the patient's status through comparison of algorithms and machine learning modules	87.7% A; 10.3% B; 2% D; 0% E
Preya Anupongongarch et al., 2022 [22]	Design and study of a prototype able to recognize blood glucose and heart rate	73.7% A; 26.7% B

Reported Clarke error grid. BG = blood glucose, PPG = photoplethysmography, MWPPG = multi wavelength photoplethysmography, EMD = empirical mode decomposition, BGL = blood glucose level, SBP = systolic blood pressure, DBP = diastolic blood pressure.

D. Innovations

MWPPG sensors represent a non-invasive optical technique that measures changes in light absorption for the estimation of arterial oxygen saturation, heart rate, blood pressure, respiratory rate, and many other parameters. These sensors are used in clinical settings to the early identification of cardiac and respiratory diseases, such as cardiac arrhythmia, respiratory failures, and chronic obstructive pulmonary diseases. The limits of the use of MWPPG sensors are related to motion artifacts and different skin pigmentation, low perfusion, venous pulsations, and the interference of concurrent volume changes in other vessels: these factors might introduce errors and compromise the PPG signal recorded. In this section we have collected eight different studies in which innovative approaches and technologies were used to improve the measurement of PPG signals with multi-wavelength PPG sensors and to explore their use in innovative positions.

The first two studies provide innovative methods for a broader assessment of blood perfusion [24], by measuring changes in hemoglobin concentrations from the continuous (DC) component of the PPG signal, and a depth resolution of skin blood pulsation [25], by exploiting the wavelength dependence of light penetration in skin tissue with a multi-wavelength multi-layer light-skin interaction model derived from the modified Beer-Lambert law. In both cases a multi-wavelength PPG sensor composed by a reflectance PPG

TABLE VIII
RESULTS OF MWPPG STUDIES WITH INNOVATIVE APPROACHES

	Wavelengths	Reference Signal	Accuracy	Correlation Coefficient (r^2)
Abay & Kyriacou, 2015 [24]	660nm 880nm	Conventional pulse oximeter, NIRS system	Forearm (B): 92.0 ± 3.1% Finger (B): 98.5 ± 0.8% Forearm (VO): 98.9 ± 2.6% Finger (VO): 88.9 ± 3.6% Forearm (TO): 78.3 ± 1.8% Finger (TO): 73.7 ± 12.8%	(ZenPPG and NIRS, p<0.05) HHb: 0.960 (VO) 0.988 (TO) HbO2: 0.821 (VO) 0.940 (TO) tHb: 0.974 (VO) 0.938 (TO)
Liu et al., 2016 [25]	470nm 570nm 590nm 940nm	Non-invasive blood pressure measurement, ECG signal	Mean absolute difference: 5.7 mmHg (IR_PPG) 4.0 mmHg (D2_PPG) 2.9 mmHg (D3_PPG)	(reference and estimated SBP, p<0.05) IR_PPG: 0.86 D2_PPG: 0.92 D3_PPG: 0.95

NIRS = Near-Infrared Spectroscopy, HbO2 = oxy-hemoglobin, HHb = deoxyhemoglobin, tHb = total hemoglobin, B = baseline, VO = venous occlusion, TO = total occlusion, ZenPPG = reflectance PPG probe, ECG = electrocardiogram, SBP = systolic blood pressure, IR_PPG = single-wavelength (infrared) single-layer model, D2_PPG = two-wavelengths (infrared, green) two-layers model, D3_PPG = three-wavelengths (infrared, green, yellow) three-layers

probe and a PPG processing system was used. The main experiments' features and results are shown in Table VII & VIII.

TABLE IX
DETAILS OF MWPPG STUDIES WITH INNOVATIVE TECHNOLOGIES

	Estimated Parameters	Number of subjects	Sensor position	Measurement condition
Alharbi et al., 2019 [26]	SpO2	31 healthy volunteers divided in 2 protocols	mOEPS on the left-hand's palm	15 subjects at rest 4 different positions on the left wrist
Yan et al., 2017 [28]	Pulse Rate	33 healthy male volunteers with 5 different types of skin	mOEPS on the left-hand's palm	resting, walking, jogging and running
Zheng et al., 2023 [27]	SpO2, Pulse rate, Respiration rate	34 healthy volunteers divided in 2 protocols	mOEPS on the subject's thumb	12 subjects during cycling and walking 22 subjects during walking, running, jumping, push-ups and boxing
Ballaji et al., 2020 [29]	SpO2	10 healthy volunteers	textile sleeve in contact with the subject's wrist	4 different positions on the left wrist

Presentation of 4 papers with their estimated parameters, their number of subjects, where the sensor is located and the measurement protocol. SpO2 = arterial oxygen saturation, mOEPS = multi-wavelength optoelectronic patch sensor.

TABLE VII
DETAILS OF MWPPG STUDIES WITH INNOVATIVE APPROACHES

	Estimated Parameters	Number of subjects	Sensor position	Measurement condition
Abay & Kyriacou, 2015 [24]	SpO2, HbO2, HHb, tHb	21 healthy volunteers	Sensor placed on subject's forearm	Induction of VO and TO
Liu et al., 2016 [25]	Blood pressure	10 patients with cardiovascular diseases, 10 healthy subjects	Sensor placed on the left ring fingertip	Subjects at rest in seated position

Presentation of 2 papers with their estimated parameters, their number of subjects, where the sensor is located and the measurement protocol. SpO2 = arterial oxygen saturation, HbO2 = oxy-hemoglobin, HHb = deoxyhemoglobin, tHb = total hemoglobin, VO = venous occlusion, TO = total occlusion.

Four studies show the use of innovative technologies to improve the recording of PPG signals through the removal of motion artifacts and the reduction of the variations in different skin types. In three studies, a multi-wavelength optoelectronic patch sensor (mOEPS), made by 16 LEDs arranged as four channels of LEDs with four LEDs in each channel, was used to provide the same wavelength illumination. The sensor was placed on the palm and on the thumb of the subject in order to explore the use of green and orange wavelengths to provide PPG signals without the influence of motion artifacts [26], to remove motion artifacts by minimizing the residue between measured PPG signal and motion estimates obtained from an accelerometer [27], and to obtain reliable measurements of the heart rate independently from the subject's skin type since a suitable lighting for the right skin type is selected via an automatic adaptation module in the

mOEPS [28]. Another study has analyzed the use of 4 plastic optical fiber (POF) probes integrated into a stretchable textile sleeve to prevent displacements of the garment and reduce motion artifacts [29]. The main experiments' features and results are shown in Tables IX & X.

The last two studies explore the use of a multi-wavelength PPG sensor integrated in innovative probes to perform a continuous perfusion monitoring directly from internal tissues. A flexible dual-wavelength PPG reflectance sensor with a specific geometry and a rounded edges covering was used to provide continuous and quantitative measurements of the vitality of intestinal tissue by measuring the SpO2 of the intestinal mucosa [30] and to detect reliable PPG signals under conditions of low peripheral perfusion (anesthetized patients) by placing the sensor on an endotracheal tube [31]. The main experiments' features and results are shown in Tables XI & XII.

TABLE XI
DETAILS OF MWPPG STUDIES WITH INNOVATIVE PROBES

	Estimated Parameters	Number of subjects	Sensor position	Measurement condition
Patel et al., 2017 [30]	SpO2	12 healthy volunteers	Sensor placed in contact with a section of pig intestine, Sensor placed in the mouth against the buccal mucosa	*in-vitro* thermal study to evaluate the thermal contribution of the sensor, *in-vivo* study in the human buccal mucosa
May et al., 2019 [31]	SpO2, Pulse rate	10 anaesthetized patients	Sensor placed on an endotracheal tube sitting past the larynx near the subglottis	Signals recorded during surgery and ceased after 1 hour

Presentation of 2 papers with their estimated parameters, their number of subjects, where the sensor is located and the measurement protocol. SpO2 = arterial oxygen saturation.

TABLE XII
RESULTS OF MWPPG STUDIES WITH INNOVATIVE PROBES

	Wavelengths	Reference Signal	Accuracy	Correlation Coefficient (r^2)
Patel et al., 2017 [30]	660nm 880nm	Conventional pulse oximeter	Mean (SpO2 %) ± SD Buccal mucosa: 98.67 ± 1.07 Index finger: 99 ± 0.74	0.80 (p<0.05)
May et al., 2019 [31]	660 nm 940 nm	Clinical monitor, Commercial pulse oximeter	SpO2 ET: 97.1 ± 1.0 PER: 97.8 ± 0.4 CM: 98.7 ± 0.7 Pulse Rate (bpm) ET: 65.4 ± 10.0 PER: 65.9 ± 9.6 CM: 64.7 ± 9.9	

SpO2 = arterial oxygen saturation, SD = standard deviation, ET = endotracheal sensor, PER = peripheral sensor, CM = clinical monitor.

IV. Discussion

A. Drug Delivery Monitoring

The therapeutic agents that can be sensed by using the previous cited method are only the ones which have optically extinctive bands inside the water window: it means that the substances that can be detected must have the peak of absorption inside the region where water (the main component of blood) has its absorption's minimum, that is roughly between 200nm and 900nm; actually this is the range in which the wavelengths used in the different studies are located [9]. In addition to that, in the blood there's the presence of hemoglobin which has a high absorption below 600nm which is in the middle of the previous specified range:

TABLE X
RESULTS OF MWPPG STUDIES WITH INNOVATIVE TECHNOLOGIES

	Wavelengths	Reference Signal	Accuracy	Correlation Coefficient (r^2)
Alharbi et al., 2019 [26]	525nm 595nm 650nm 870nm	Conventional pulse oximeter		1st subprotocol: 0.9604 (p=0.88) 2nd subprotocol: 0.9604 (p=0.88)
Yan et al., 2017 [28]		ECG signal		At rest: 0.9643 (p=0.99)
Zheng et al., 2023 [27]		SpO2: conventional pulse oximeter, HR: chest strap, RR: metabolic cart	SpO2: ≥ 95% HR: 95% RR: 95%	HR: 0.9952 RR: 0.8314
Ballaji et al., 2020 [29]	660 nm 850 nm	Conventional pulse oximeter	(participant P9) (A) POF: -; Reference: 98.9 (B) POF: 98.1; Reference: 98.3 (C) POF: -; Reference: 99 (D) POF: 99.5; Reference: 99	

SpO2 = arterial oxygen saturation, HR = heart rate, RR = respiratory rate, ECG = electrocardiogram, POF = plastic optical fiber.

the sensing of agents which have their peak inside the water range, but below this threshold, can be harder but still possible [10].

With this approach the delivered dose can be verified and controlled in time, as well as its clearance properties can be quantified: all these assessments cannot be performed continuously and non-invasively using other invasive methods, that are the ones that typically involve blood collection and analysis; so, overall, this impacts on the patient safety, on the treatment quality and reproducibility [9]. Additionally, these assessments can be used to perform an intravenous administration of drugs based on the achieved plasma concentration: in this way the concentration's curve can be continuously monitored during the injection time and, with a closed loop system, the rate of injection can be changed according to the measured concentrations [13].

Unfortunately, these studies were performed only on animals, also because the experiments involved the injection of substances that could be toxic: a validation phase needs to be performed to let the health care system benefit from this innovation.

B. Microcirculation

In Tables III & IV we find the studies we've reviewed on this topic. In the experiments, a specific wavelength could turn out to be the most suitable for extracting parameters of interest, that is why, after having performed some evaluations, only a smaller range within the whole spectrum may be considered in further investigations.

Green appears the preferable wavelength; in fact, in microcirculation, the most relevant wavelengths are usually the shortest ones due to their inability to penetrate deep into the skin. The result is that the green light focuses on the capillary bed without reaching the arteries, so it's unable to capture the main vessels' pulsatile activity. However, it has been hypothesized that the capillaries, since continuously compressed and decompressed by arterial pressure, could reflect the behavior of the deepest layer, what is more likely to be true especially during heating tests or after applying pressure. What is more, this wavelength is less affected by motion artifacts and it has a better chance of extracting pulse frequency thanks to the correlation with ECG; that's why, after having used different sources of light, it has found its place in the evaluation of sublingual microcirculation [13], for which the tissue perfusion calculated could work like a mirror of visceral microcirculation, without incurring in invasive measurements. Note that this wavelength is mainly absorbed by melanin, which is less present on tongue. Also a separate study [15], which is based on iPPG and 2 different wavelengths (green: 530nm and IR: 810nm), shows that during local heating test as well as during liniment application the perfusion changes measured by the PPGgreen are correlated with the ones extracted by the gold standard LDI (laser Doppler imaging); even in the other two tests here described, the post-occlusive reactive hyperemia test and the regional anesthesia test, the correlation with the results from LDI is shared by both the green and the IR wavelengths. In the calculation of BPA [17], defined here with AC-to-DC ratio, the green choice is again the winning one, since its SNR ratio is the highest. The BPA positively correlates with Temperature, in the case of a pre-cooled finger, it has been shown the importance of considering the delay with which BPA increases as the finger starts heating; in fact, this parameter is shared among the different subjects. This iPPG could be used to show the assessment of the changes in cutaneous blood linked to thermoregulation, by revealing the presence of cardiovascular complications or all kinds of abnormality that affect the Temperature.

Applying an increasing external pressure to the patient's finger it's possible to appreciate how the variability of the absorption distributions is mainly allocated to the shortest wavelengths; at low pressure applied at the beginning, the three shorter-wavelength channels (465nm blue, 515nm green, 590 nm yellow) are related to the superficial vessels, as the external pressure increases they can focus on the deepest layers, for example the yellow channel reveals a larger contribution of its signal from the deeper vessels than the red channel, which has been the most related to arteries by far. This is also confirmed by performing oscillometric measurements under an increasing pressure, thanks to which the maximum oscillometric pressure "MOP" could be extracted by the signals, this parameter is different from the traditional mean arterial pressure "MAP". Under no external pressure, the value of the MOPs decreases with the wavelength, so with the poor capability of penetration, what agrees with the principle of the systemic circulation: blood pressure decreases as vessels move from arteries to arterioles, capillaries, venules and veins. Note that the infrared channel and the pressure obtained by the brachial cuff (Omron M3) are correlated, since they are both linked to the MAP [16].

As seen, some studies include the iPPG, so imaging photoplethysmography, and one is about remote PPG; the latter is associated with a single wavelength only (peak at 760 nm), despite that, the focus on processing algorithms, capable of creating distribution maps based on microcirculation parameters, results in something interesting. Regarding that, to assess the perfusion in the patient's hand after the regional anesthesia the comparison between rPPG maps with the most common thermal images relies on a qualitative

observation of the different perfusion maps obtained by modifying the attenuation value p: a higher p takes place alongside the sensitivity of RA detection, the risk of increasing this parameter consists in receiving an over-blurred map image [14].

In conclusion, microcirculation can hide fundamental shreds of evidence concerning the state of health of a person, that's why analyzing the MWPPG and finding the best combination of wavelengths according to the clinical examinations should be useful not necessarily to overcome the traditional devices, but as support to other tools.

C. Blood Glucose Monitoring

The visible light (VIS), typically red and green light, allows us to reach the dermis tissue, in particular arterioles and capillaries region, while the near infrared (NIR) spectrum can reach also arteries.

As demonstrated in the study [18] took in consideration, to study the blood glucose level, the use of a Vis-NIR spectrum is a better solution than the use of a Vis-spectrum or a Nir-spectrum based model, providing a more precise depth resolution of blood pulsation under the wrist. Emitting Vis-NIR light to the arteries, it's possible to obtain pulsed-periodic signals, related to the blood volume change in the vessel.

Before signal processing, which provides the digital version of PPG signal, it is required to remove from the raw signal the noise artifacts, in particular moving artifacts and baseline drifts, otherwise there could problems in true blood glucose (BG) recognition. The pre-processing phase could consist in current to voltage converter, high pass filter, non-inverting amplifier and low pass filter. A complete processing algorithm is composed by: pre-processing phase, peak detection, signal segmentation, feature extraction and the built of a prediction model.

A solution, in the building of a system able to detect BG level, as suggested by [21] study, consists in analyzing the effect on the shape of PPG waveform of physiological alterations. The procedure of pre-processing in this case starts from the PPG signal, which is digitalized and then pass through an activity detection module. The output of this module is a vector containing a set of features, this vector can be seen as signal or no signal. This vector is computed for each frame, and it's also derived from the heart rate, the oxygen saturation range, from information about the individual (weight, body mass, age, body index) and spectral entropy statistics. Differently, in the study [20] after the processing are obtained features from the PPG signal. It was also performed Empirical Mode Decomposition (EMD), a technique to decompose the signal in basic parts called IMFs. PPG-waveform based, and IMFs-based features are used as input for the regression algorithm to test machine learning techniques. A machine learning technique works better not just considering a large amount of features, but also consider an importance-based selection of the features. The study [20] calculated a total of 420 IMFs-based features for three wavelengths, but only 30 were selected for the use. 40 individuals participated at [23] study, the model for the BG level evaluation is based on a BPNN (Back Propagation Neural Network) to whose were given as input 12 x 40 statistical features, 11 x 40 bioelectric impedance features and 7 x 40 physiological features of the volunteers, the total features are then weighted.

Non-invasive devices based on PPG for estimating BG levels may have great potential also for clinical implications but can still be improved as suggested in the paper [22], in fact the [20] study aims to collaborate with medical institutions to acquire more data and improve the performance of the method. [22] acquired data putting the finger into the probe and reveals that in their study that arm position, nail paints, thick nails, blue fingernails due to congenital disease, finger larger than the probe affects the accuracy. Furthermore, this study aims to improve the performances in the BG range > 150 mg/dL.

D. Innovations

The use of a PPG reflectance probe on the forearm [24] shows a good accuracy for the measurement of SpO2 and a high correlation between PPG-derived and NIRS-derived hemoglobin's concentrations. While SpO2 from the fingers did not exhibit any considerable mean desaturation from baseline during venous occlusion, SpO2 from the forearm showed a statistically significant desaturation with a mean drop of 3.1±3.6%, as expected. However, the baseline SpO2 estimated by the probe presented lower values compared with SpO2 calculated from the fingertips because of the difficulty to extract high amplitude PPG from the forearm, which has a low capillary density, but also due to a higher absorption of red light in the forearm and the presence of large superficial veins on the forearm. The use of a multi-wavelength multi-layer light-skin interaction model [25] has significantly improved the measurement accuracy of blood pressure, reducing the mean absolute difference between the reference and the estimated systolic blood pressure (SBP). On most of the subjects, the IR/green light combination and the IR/yellow/green light combination were the optimal wavelength combinations for the 2-wavelenght 2-layer model and the 3-wavelenght 3-layer model respectively, but they can be different in other test conditions due to variation in signal quality, measurement location and properties of the skin tissue. The 3-layer model outperformed compared to the 2-layer model, but the computational process is more

complex so there may be a tradeoff between accuracy and simplicity.

The innovative technologies to improve the recording of PPG signals through the removal of motion artifacts and the reduction of the variations in different skin types have shown a good accuracy and high correlation coefficients with respect to the reference signals. The measurements of SpO2 with the mOEPS [26], [27] were less affected by small movements compared to those with the commercial pulse oximeter. In particular, the green and orange illuminations allow to reduce motion artifacts, making a better extraction of SpO2 values: at these wavelengths, the penetration into the skin is significantly shallower compared with the commonly used red and infrared illuminations, so the use of green and orange illuminations for SpO2 can work during free hand movement and physical activity. The results from the mOEPS with an automatic adaptation module [28] show that there is no significant difference in the 5 groups tested with different skin types during 4 types of exercises: this technology can be used to obtain reliable measurements of the heart rate (HR) independently from the subject's skin type. While resting and moderate physical activity show precise HR measurements, as the intensity of the physical activity increased, the intensity of the PPG signals corresponding to the illuminations of yellow, red and infrared LEDs was also increased significantly, whilst the intensity of the PPG signals from the green illumination remained stable: this confirms that green illumination isn't influenced by motion artifacts. The use of a sleeve textile with four plastic optical fiber (POF) probes [29] shows that at least one pair of fibers (out of 4) allows the PPG signal to be detected with a good accuracy and with a mean difference of SpO2 equal to –0.03% between the developed sensor and a commercial oximeter. The lack of a PPG signal in the other fibers is a positional effect, not a light collection issue, so it is related to whether the sensor doesn't lie near a major vessel.

The results of the two studies [30] [31], in which the multi-wavelength PPG sensor has been integrated in innovative probes, show a very high accuracy in measuring SpO2 and pulse rate directly from internal tissues, with two wavelengths (red and infrared). In both studies, the PPG signals recorded by the probes were of good quality and large amplitudes, with a high signal-to-noise ratio: 23.4 dB and 35.6 dB from the buccal mucosa [30] and 17.7 dB and 23.7 dB from the trachea [31], for red and infrared signals respectively. As expected, by using the sensor from the inside, such as near the subglottis for the endotracheal sensor and near the buccal mucosa for the intraluminal sensor, we are closer to the network of blood vessels and so we have a richer blood supply and a better perfusion. Despite these positive aspects, the newly developed intraluminal intestinal PPG sensor was used to measure SpO2 in the buccal mucosa, not in the intestinal mucosa, so further clinical studies must be done to validate the intraluminal intestinal PPG sensor in human large intestine. Moreover, the variance of the PPG signals recorded by the endotracheal sensors from the trachea was greater than the one in the periphery while the signal-to-noise was lower (19.8 dB and 27.5 dB from the periphery for red and infrared respectively): this suggests that either the trachea is more susceptible to variations in blood supply caused by sympathetic responses, or that the sensor itself may not have been adequately located during the placement procedure, as there was no way to confirm the sensor location after placement.

This review focused on some of the most innovative applications of multi-wavelength photoplethysmography in monitoring physiological parameters. However, there are other applications in the literature that were not explored in this work, which also deserve more in-depth study.

V. Conclusion

The potential applications of MWPPG sensors in monitoring physiological parameters are numerous. In this review, we have aimed to delve into the most innovative and promising ones in the field of scientific research.

The use of MWPPG sensors ensures continuous glucose measurement with a delay of only a few minutes and can be integrated into devices like smartwatches, making them usable as low-cost, completely non-invasive tools for prevention [18]. Research on this type of device aims to reduce the error in acquiring blood glucose values in hyperglycemic or hypoglycemic patients [22], decrease the output delay to seconds instead of minutes [21], and select deep learning tools to improve performance [20]. The goal is to employ devices based on this technology in clinical settings as well.

Moreover, MWPPG sensors can be applied to detect the presence of a wide range of optically active drugs and nanoparticles within blood vessels. This is particularly useful as it allows for knowledge of the initial and maintenance dose of these substances, as well as their circulation characteristics and therapeutic window [9],[7].

Another promising field for the application of MWPPG sensor technology is microcirculation monitoring. This application allows for higher resolution in PIV (pressure induced vasodilatation) measurements [16], enabling the study of vasomotor activity effects at various depths of skin vascularization and the calculation of pressures at the levels of large and small arteries, thus aiding in the prevention of hypertension [16]. Additionally, this type of application allows for more accurate heart rate monitoring due to its low sensitivity to motion artifacts and is considered very interesting and widely applicable, as it can detect the

presence of conditions such as diabetes mellitus, neurological diseases, and rheumatological diseases with a high risk of cardiovascular complications [17].

In exploring the major innovations proposed for monitoring "classical" physiological parameters (e.g., oxygen saturation and heart rate) using MWPPG sensors, it became evident that the trend in this scientific field is to develop miniaturized devices or devices that can be worn by patients in everyday life and during sports activities without being a hindrance [24]. The goal is to create devices that, with a high level of accuracy, can provide diagnoses and prognoses for various cardiovascular and respiratory conditions and can effectively address the issue of motion artifacts through appropriate algorithms [27]. These features could also lead to the application of these devices in monitoring the physiological parameters of fetuses or newborns [29]. However, a challenge being studied in the application of these sensors is their usability in individuals with very different skin compositions and pigmentations [28]. Another research avenue in this field is the search for alternative placements for MWPPG sensors. Promising studies have been conducted to investigate their applicability in the intestinal lumen and the trachea, but much work remains to be done to validate them [30],[31].

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